

CABLE THEFT & VANDALISM

MONITORING & SURVEILLANCE SYSTEM

By JSNM group

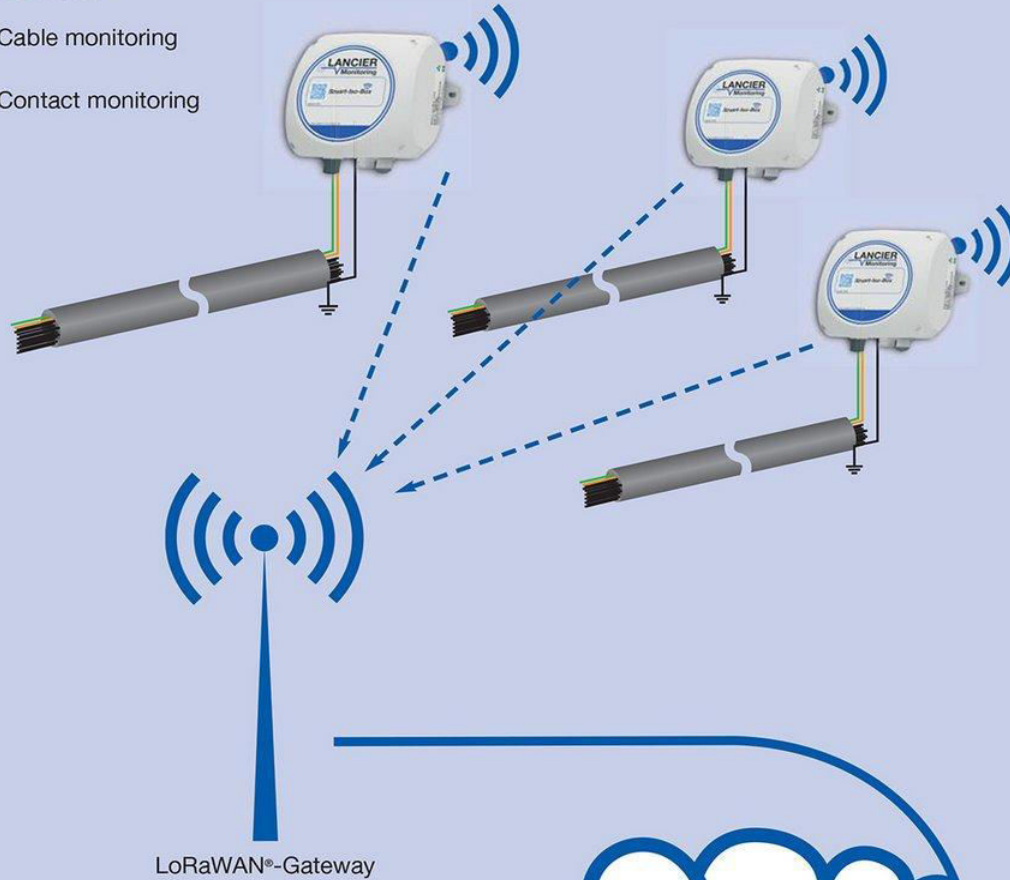
SUMMARY

The proposed Cable Theft and Vandalism Monitoring and Surveillance System combines IoT devices, GPS tracking, geo fencing, acoustic sensors, AI analytics, and a centralized database. IoT sensors continuously monitor protected infrastructure and transmit real-time data to the server. GPS and geo fencing detect unauthorized movement or access to restricted areas, while acoustic sensors identify sounds associated with cable theft. AI analyzes the collected data to reduce false alarms and identify high-risk events. The processed information is displayed on a dashboard, enabling security personnel to respond quickly and improving the protection of critical infrastructure.

System diagramme

Smart-Iso-Box 200/10

- 1 x Cable monitoring
- 2 x Contact monitoring



Evaluation of the measurement data and alarm report with visualisation of its geographical position in the LANCER Monitoring UMS System.

INTRODUCTION

- A **Cable Theft and Vandalism Monitoring and Surveillance System** is a system designed to help organizations such as power utilities, railway operators, municipalities, and telecommunications companies monitor, report, and respond to incidents of cable theft and infrastructure vandalism.

IoT DASHBOARD DEVELOPMENT

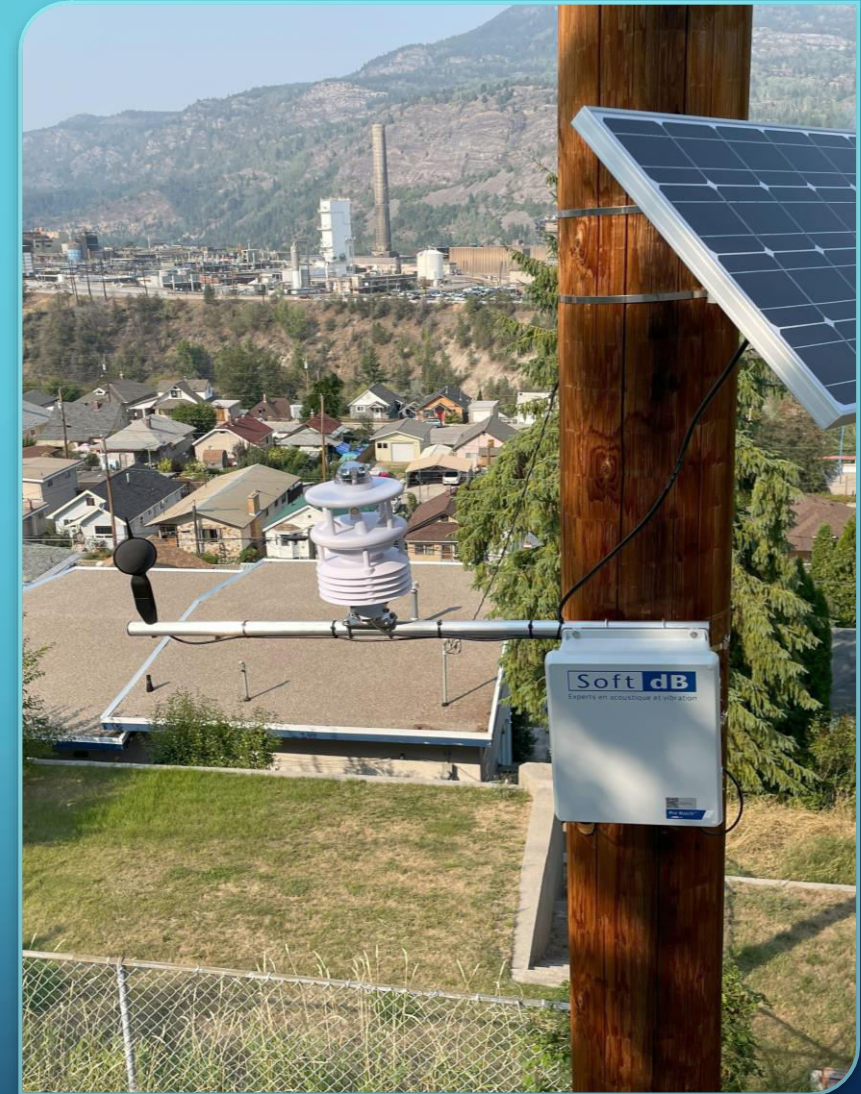
An **IoT dashboard** is the central screen where users monitor all connected devices and incidents in real time. It displays information received from sensors installed on electrical cables, substations, and other critical infrastructure. It can display: Number of active sensors, Live GPS locations, Active theft or vandalism alerts, CCTV status, Maps showing protected areas



ACOUSTIC SENSORY FRAMEWORK

Examples of sounds the system can recognize include:

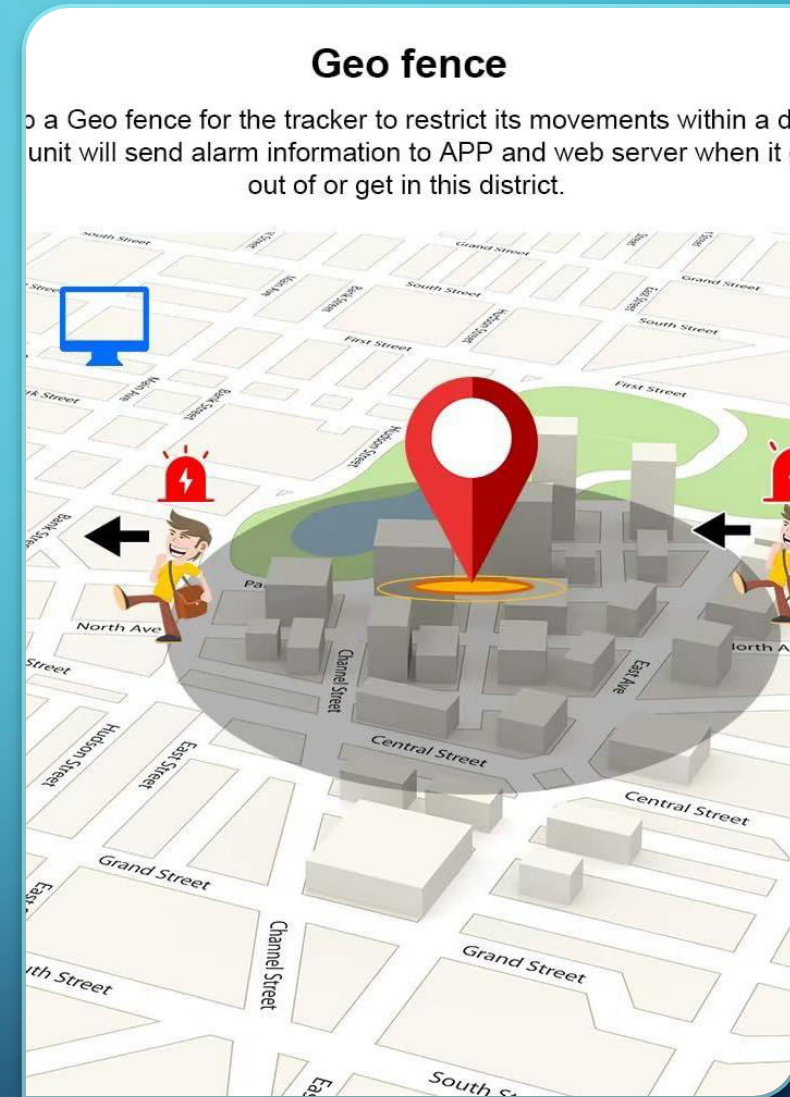
- Metal cutting
- Hammering
- Drilling
- Digging
- Cable pulling
- Explosions



GEOFENCING

When someone enters a restricted zone, your application could:

- Display a warning on the dashboard.
- Save the incident in the database.
- Notify security personnel.
- Mark the incident as requiring immediate attention



AI ANALYTICS

Artificial Intelligence (AI) analyzes data collected from sensors to identify suspicious activity and predict risks. Instead of simply detecting noise, it can distinguish between: Wind, Rain, Passing vehicles, Metal cutting, Digging, Hammering.

AI can also: Predict theft hotspots on historical incidents, Detect abnormal sensor behavior, Prioritize high-risk alerts and Reduce false alarms

